SG No. 124 D_{4h}^2 $P4/mcc$ [tetragonal]

* plus set: $+[0, 0, 0]$
* Wyckoff site: 2a, site symmetry: 422

Table 1: Wyckoff bond: 2a@2a, bond symmetry: 422

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, \frac{1}{4}]$	[1,2,3,4,-5,-6,-7,-8]
2	$[0, 0, -Z]$	$[0, 0, \frac{3}{4}]$	[9,10,11,12,-13,-14,-15,-16]

Table 2: Wyckoff bond: 4b@2a, bond symmetry: 222.

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, 0, \frac{1}{4}]$	[1,-2,-5,6]
2	$[0, X, 0]$	$[0, 0, \frac{1}{4}]$	[3,-4,7,-8]
3	$[-X, 0, 0]$	$[0, 0, \frac{3}{4}]$	[9,-10,-13,14]
4	$[0, -X, 0]$	$[0, 0, \frac{3}{4}]$	[11,-12,15,-16]

Table 3: Wyckoff bond: 4c@2a, bond symmetry: 2.22

No.	vector	center	mapping
1	$[X, X, 0]$	$[0, 0, \frac{1}{4}]$	[1,-2,7,-8]
2	$[-X, X, 0]$	$[0, 0, \frac{1}{4}]$	[3,-4,5,-6]
3	$[-X, -X, 0]$	$[0, 0, \frac{3}{4}]$	[9,-10,15,-16]
4	$[X, -X, 0]$	$[0, 0, \frac{3}{4}]$	[11,-12,13,-14]

Table 4: Wyckoff bond: 8d@2a, bond symmetry: .2.

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, 0, \frac{1}{4}]$	[1,-5]
2	$[-X, 0, Z]$	$[0, 0, \frac{1}{4}]$	[2,-6]
3	$[0, X, Z]$	$[0, 0, \frac{1}{4}]$	[3,-8]
4	$[0, -X, Z]$	$[0, 0, \frac{1}{4}]$	[4,-7]
5	$[-X, 0, -Z]$	$[0, 0, \frac{3}{4}]$	[9,-13]
6	$[X, 0, -Z]$	$[0, 0, \frac{3}{4}]$	[10,-14]
7	$[0, -X, -Z]$	$[0, 0, \frac{3}{4}]$	[11,-16]
8	$[0, X, -Z]$	$[0, 0, \frac{3}{4}]$	[12,-15]

Table 5: Wyckoff bond: 8e@2a, bond symmetry: ..2

No.	vector	center	mapping
1	$[X, X, Z]$	$[0, 0, \frac{1}{4}]$	[1,-8]
2	$[-X, -X, Z]$	$[0, 0, \frac{1}{4}]$	[2,-7]
3	$[-X, X, Z]$	$[0, 0, \frac{1}{4}]$	[3,-6]
4	$[X, -X, Z]$	$[0, 0, \frac{1}{4}]$	[4,-5]

continued ...

Table 5

No.	vector	center	mapping
5	$[-X, -X, -Z]$	$\left[0, 0, \frac{3}{4}\right]$	`[9,-16]`
6	$[X, X, -Z]$	$\left[0, 0, \frac{3}{4}\right]$	`[10,-15]`
7	$[X, -X, -Z]$	$\left[0, 0, \frac{3}{4}\right]$	`[11,-14]`
8	$[-X, X, -Z]$	$\left[0, 0, \frac{3}{4}\right]$	`[12,-13]`

Table 6: Wyckoff bond: `8f@2a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, 0, \frac{1}{4}\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[0, 0, \frac{1}{4}\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[0, 0, \frac{1}{4}\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[0, 0, \frac{1}{4}\right]$	`[7,-8]`
5	$[-X, -Y, 0]$	$\left[0, 0, \frac{3}{4}\right]$	`[9,-10]`
6	$[Y, -X, 0]$	$\left[0, 0, \frac{3}{4}\right]$	`[11,-12]`
7	$[X, -Y, 0]$	$\left[0, 0, \frac{3}{4}\right]$	`[13,-14]`
8	$[-Y, -X, 0]$	$\left[0, 0, \frac{3}{4}\right]$	`[15,-16]`

Table 7: Wyckoff bond: `16g@2a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, 0, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, 0, \frac{1}{4}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[0, 0, \frac{1}{4}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[0, 0, \frac{1}{4}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[0, 0, \frac{1}{4}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[0, 0, \frac{1}{4}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[0, 0, \frac{1}{4}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[0, 0, \frac{1}{4}\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[0, 0, \frac{3}{4}\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[0, 0, \frac{3}{4}\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[0, 0, \frac{3}{4}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[0, 0, \frac{3}{4}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[0, 0, \frac{3}{4}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[0, 0, \frac{3}{4}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[0, 0, \frac{3}{4}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[0, 0, \frac{3}{4}\right]$	`[16]`

* Wyckoff site: `2b`, site symmetry: `4/m..`

Table 8: Wyckoff bond: `2a@2b`, bond symmetry: `4/m..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, 0]$	`[1,2,3,4,-9,-10,-11,-12]`
2	$[0, 0, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[5,6,7,8,-13,-14,-15,-16]`

Table 9: Wyckoff bond: `4b@2b`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, 0]$	`[1,-2,-9,10]`
2	$[-Y, X, 0]$	$[0, 0, 0]$	`[3,-4,-11,12]`
3	$[-X, Y, 0]$	$[0, 0, \frac{1}{2}]$	`[5,-6,-13,14]`
4	$[Y, X, 0]$	$[0, 0, \frac{1}{2}]$	`[7,-8,-15,16]`

Table 10: Wyckoff bond: `8c@2b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1,-9]`
2	$[-X, -Y, Z]$	$[0, 0, 0]$	`[2,-10]`
3	$[-Y, X, Z]$	$[0, 0, 0]$	`[3,-11]`
4	$[Y, -X, Z]$	$[0, 0, 0]$	`[4,-12]`
5	$[-X, Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[5,-13]`
6	$[X, -Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[6,-14]`
7	$[Y, X, -Z]$	$[0, 0, \frac{1}{2}]$	`[7,-15]`
8	$[-Y, -X, -Z]$	$[0, 0, \frac{1}{2}]$	`[8,-16]`

* Wyckoff site: `2c`, site symmetry: `422`

Table 11: Wyckoff bond: `2a@2c`, bond symmetry: `422`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}]$	`[1,2,3,4,-5,-6,-7,-8]`
2	$[0, 0, -Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}]$	`[9,10,11,12,-13,-14,-15,-16]`

Table 12: Wyckoff bond: `4b@2c`, bond symmetry: `222.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}]$	`[1,-2,-5,6]`
2	$[0, X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}]$	`[3,-4,7,-8]`
3	$[-X, 0, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}]$	`[9,-10,-13,14]`
4	$[0, -X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}]$	`[11,-12,15,-16]`

Table 13: Wyckoff bond: `4c@2c`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}]$	`[1,-2,7,-8]`
2	$[-X, X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}]$	`[3,-4,5,-6]`
3	$[-X, -X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}]$	`[9,-10,15,-16]`
4	$[X, -X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}]$	`[11,-12,13,-14]`

Table 14: Wyckoff bond: `8d@2c`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[1,-5]`
2	$[-X, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[2,-6]`
3	$[0, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[3,-8]`
4	$[0, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[4,-7]`
5	$[-X, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[9,-13]`
6	$[X, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[10,-14]`
7	$[0, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[11,-16]`
8	$[0, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[12,-15]`

Table 15: Wyckoff bond: `8e@2c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[1,-8]`
2	$[-X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[2,-7]`
3	$[-X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[3,-6]`
4	$[X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[4,-5]`
5	$[-X, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[9,-16]`
6	$[X, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[10,-15]`
7	$[X, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[11,-14]`
8	$[-X, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[12,-13]`

Table 16: Wyckoff bond: `8f@2c`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[7,-8]`
5	$[-X, -Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[9,-10]`
6	$[Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[11,-12]`
7	$[X, -Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[13,-14]`
8	$[-Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[15,-16]`

Table 17: Wyckoff bond: `16g@2c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[7]`

continued ...

Table 17

No.	vector	center	mapping
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[16]`

* Wyckoff site: `2d`, site symmetry: `4/m..`

Table 18: Wyckoff bond: `2a@2d`, bond symmetry: `4/m..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,2,3,4,-9,-10,-11,-12]`
2	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6,7,8,-13,-14,-15,-16]`

Table 19: Wyckoff bond: `4b@2d`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,-2,-9,10]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-4,-11,12]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,-6,-13,14]`
4	$[Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[7,-8,-15,16]`

Table 20: Wyckoff bond: `8c@2d`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,-9]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,-10]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-11]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,-12]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,-13]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[6,-14]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[7,-15]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[8,-16]`

* Wyckoff site: `4e`, site symmetry: `2/m..`

Table 21: Wyckoff bond: `4a@4e`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,-2,-9,10]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[3,-4,-11,12]`

continued ...

Table 21

No.	vector	center	mapping
3	$[-X, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[5,-6,-13,14]`
4	$[Y, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[7,-8,-15,16]`

Table 22: Wyckoff bond: `4b@4e`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,-9,-10]`
2	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[3,4,-11,-12]`
3	$[0, 0, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6,-13,-14]`
4	$[0, 0, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[7,8,-15,-16]`

Table 23: Wyckoff bond: `8c@4e`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,-9]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-10]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[3,-11]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,-12]`
5	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[5,-13]`
6	$[X, -Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[6,-14]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[7,-15]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[8,-16]`

* Wyckoff site: `4f`, site symmetry: `222.`

Table 24: Wyckoff bond: `4a@4f`, bond symmetry: `222.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,-2,-5,6]`
2	$[0, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[3,-4,7,-8]`
3	$[-X, 0, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[9,-10,-13,14]`
4	$[0, -X, 0]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[11,-12,15,-16]`

Table 25: Wyckoff bond: `4b@4f`, bond symmetry: `222.`

No.	vector	center	mapping
1	$[0, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,-2,5,-6]`
2	$[-X, 0, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[3,-4,-7,8]`
3	$[0, -X, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[9,-10,13,-14]`
4	$[X, 0, 0]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[11,-12,-15,16]`

Table 26: Wyckoff bond: `4c@4f`, bond symmetry: `222.`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,2,-5,-6]`
2	$[0,\ 0,\ Z]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[3,4,-7,-8]`
3	$[0,\ 0,\ -Z]$	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,10,-13,-14]`
4	$[0,\ 0,\ -Z]$	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[11,12,-15,-16]`

Table 27: Wyckoff bond: `8d@4f`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,-5]`
2	$[-X,\ 0,\ Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[2,-6]`
3	$[0,\ X,\ Z]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[3,-8]`
4	$[0,\ -X,\ Z]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[4,-7]`
5	$[-X,\ 0,\ -Z]$	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,-13]`
6	$[X,\ 0,\ -Z]$	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[10,-14]`
7	$[0,\ -X,\ -Z]$	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[11,-16]`
8	$[0,\ X,\ -Z]$	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[12,-15]`

Table 28: Wyckoff bond: `8e@4f`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0,\ X,\ Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,-6]`
2	$[0,\ -X,\ Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[2,-5]`
3	$[-X,\ 0,\ Z]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[3,-7]`
4	$[X,\ 0,\ Z]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[4,-8]`
5	$[0,\ -X,\ -Z]$	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,-14]`
6	$[0,\ X,\ -Z]$	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[10,-13]`
7	$[X,\ 0,\ -Z]$	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[11,-15]`
8	$[-X,\ 0,\ -Z]$	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[12,-16]`

Table 29: Wyckoff bond: `8f@4f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,-2]`
2	$[-Y,\ X,\ 0]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[3,-4]`
3	$[-X,\ Y,\ 0]$	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[5,-6]`
4	$[Y,\ X,\ 0]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[7,-8]`
5	$[-X,\ -Y,\ 0]$	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,-10]`
6	$[Y,\ -X,\ 0]$	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[11,-12]`
7	$[X,\ -Y,\ 0]$	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[13,-14]`
8	$[-Y,\ -X,\ 0]$	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[15,-16]`

Table 30: Wyckoff bond: `16g@4f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[16]`

* Wyckoff site: `4g`, site symmetry: `4..`

Table 31: Wyckoff bond: `4a@4g`, bond symmetry: `4..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, z]$	`[1,2,3,4]`
2	$[0, 0, -Z]$	$\left[0, 0, \frac{1}{2} - z\right]$	`[5,6,7,8]`
3	$[0, 0, -Z]$	$[0, 0, -z]$	`[9,10,11,12]`
4	$[0, 0, Z]$	$\left[0, 0, z + \frac{1}{2}\right]$	`[13,14,15,16]`

Table 32: Wyckoff bond: `8b@4g`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, z]$	`[1,-2]`
2	$[-Y, X, 0]$	$[0, 0, z]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[0, 0, \frac{1}{2} - z\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[0, 0, \frac{1}{2} - z\right]$	`[7,-8]`
5	$[-X, -Y, 0]$	$[0, 0, -z]$	`[9,-10]`
6	$[Y, -X, 0]$	$[0, 0, -z]$	`[11,-12]`
7	$[X, -Y, 0]$	$\left[0, 0, z + \frac{1}{2}\right]$	`[13,-14]`
8	$[-Y, -X, 0]$	$\left[0, 0, z + \frac{1}{2}\right]$	`[15,-16]`

Table 33: Wyckoff bond: `16c@4g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, z]$	`[1]`
2	$[-X, -Y, Z]$	$[0, 0, z]$	`[2]`

continued ...

Table 33

No.	vector	center	mapping
3	$[-Y, X, Z]$	$[0, 0, z]$	`[3]`
4	$[Y, -X, Z]$	$[0, 0, z]$	`[4]`
5	$[-X, Y, -Z]$	$\left[0, 0, \frac{1}{2} - z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[0, 0, \frac{1}{2} - z\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[0, 0, \frac{1}{2} - z\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[0, 0, \frac{1}{2} - z\right]$	`[8]`
9	$[-X, -Y, -Z]$	$[0, 0, -z]$	`[9]`
10	$[X, Y, -Z]$	$[0, 0, -z]$	`[10]`
11	$[Y, -X, -Z]$	$[0, 0, -z]$	`[11]`
12	$[-Y, X, -Z]$	$[0, 0, -z]$	`[12]`
13	$[X, -Y, Z]$	$\left[0, 0, z + \frac{1}{2}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[0, 0, z + \frac{1}{2}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[0, 0, z + \frac{1}{2}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[0, 0, z + \frac{1}{2}\right]$	`[16]`

* Wyckoff site: `4h`, site symmetry: `4..`

Table 34: Wyckoff bond: `4a@4h`, bond symmetry: `4..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1,2,3,4]`
2	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[5,6,7,8]`
3	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[9,10,11,12]`
4	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[13,14,15,16]`

Table 35: Wyckoff bond: `8b@4h`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[7,-8]`
5	$[-X, -Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[9,-10]`
6	$[Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[11,-12]`
7	$[X, -Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[13,-14]`
8	$[-Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[15,-16]`

Table 36: Wyckoff bond: `16c@4h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[4]`

continued ...

Table 36

No.	vector	center	mapping
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[16]`

* Wyckoff site: `8i`, site symmetry: `2..`

Table 37: Wyckoff bond: `8a@8i`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, 0, z\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	`[7,-8]`
5	$[-X, -Y, 0]$	$\left[0, \frac{1}{2}, -z\right]$	`[9,-10]`
6	$[Y, -X, 0]$	$\left[\frac{1}{2}, 0, -z\right]$	`[11,-12]`
7	$[X, -Y, 0]$	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	`[13,-14]`
8	$[-Y, -X, 0]$	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	`[15,-16]`

Table 38: Wyckoff bond: `8b@8i`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1,2]`
2	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[3,4]`
3	$[0, 0, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	`[5,6]`
4	$[0, 0, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	`[7,8]`
5	$[0, 0, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	`[9,10]`
6	$[0, 0, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[11,12]`
7	$[0, 0, Z]$	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	`[13,14]`
8	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	`[15,16]`

Table 39: Wyckoff bond: `16c@8i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[2]`

continued ...

Table 39

No.	vector	center	mapping
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, 0, z\right]$	[3]
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, 0, z\right]$	[4]
5	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	[5]
6	$[X, -Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	[6]
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	[7]
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	[8]
9	$[-X, -Y, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	[9]
10	$[X, Y, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	[10]
11	$[Y, -X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	[11]
12	$[-Y, X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	[12]
13	$[X, -Y, Z]$	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	[13]
14	$[-X, Y, Z]$	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	[14]
15	$[-Y, -X, Z]$	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	[15]
16	$[Y, X, Z]$	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	[16]

* Wyckoff site: 8j, site symmetry: ..2

Table 40: Wyckoff bond: 8a@8j, bond symmetry: ..2

No.	vector	center	mapping
1	$[X, -X, Z]$	$\left[x, x, \frac{1}{4}\right]$	[1,-7]
2	$[-X, X, Z]$	$\left[-x, -x, \frac{1}{4}\right]$	[2,-8]
3	$[X, X, Z]$	$\left[-x, x, \frac{1}{4}\right]$	[3,-5]
4	$[-X, -X, Z]$	$\left[x, -x, \frac{1}{4}\right]$	[4,-6]
5	$[-X, X, -Z]$	$\left[-x, -x, \frac{3}{4}\right]$	[9,-15]
6	$[X, -X, -Z]$	$\left[x, x, \frac{3}{4}\right]$	[10,-16]
7	$[-X, -X, -Z]$	$\left[x, -x, \frac{3}{4}\right]$	[11,-13]
8	$[X, X, -Z]$	$\left[-x, x, \frac{3}{4}\right]$	[12,-14]

Table 41: Wyckoff bond: 8b@8j, bond symmetry: ..2

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[x, x, \frac{1}{4}\right]$	[1,7]
2	$[-X, -X, 0]$	$\left[-x, -x, \frac{1}{4}\right]$	[2,8]
3	$[-X, X, 0]$	$\left[-x, x, \frac{1}{4}\right]$	[3,5]
4	$[X, -X, 0]$	$\left[x, -x, \frac{1}{4}\right]$	[4,6]
5	$[-X, -X, 0]$	$\left[-x, -x, \frac{3}{4}\right]$	[9,15]
6	$[X, X, 0]$	$\left[x, x, \frac{3}{4}\right]$	[10,16]
7	$[X, -X, 0]$	$\left[x, -x, \frac{3}{4}\right]$	[11,13]
8	$[-X, X, 0]$	$\left[-x, x, \frac{3}{4}\right]$	[12,14]

Table 42: Wyckoff bond: `16c@8j`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, x, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, -x, \frac{1}{4}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[-x, x, \frac{1}{4}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[x, -x, \frac{1}{4}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, x, \frac{1}{4}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x, -x, \frac{1}{4}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[x, x, \frac{1}{4}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[-x, -x, \frac{1}{4}\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[-x, -x, \frac{3}{4}\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[x, x, \frac{3}{4}\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[x, -x, \frac{3}{4}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[-x, x, \frac{3}{4}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[x, -x, \frac{3}{4}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[-x, x, \frac{3}{4}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[-x, -x, \frac{3}{4}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[x, x, \frac{3}{4}\right]$	`[16]`

* Wyckoff site: `8k`, site symmetry: `.2.`

Table 43: Wyckoff bond: `8a@8k`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, X, Z]$	$\left[x, 0, \frac{1}{4}\right]$	`[1,-6]`
2	$[0, -X, Z]$	$\left[-x, 0, \frac{1}{4}\right]$	`[2,-5]`
3	$[-X, 0, Z]$	$\left[0, x, \frac{1}{4}\right]$	`[3,-7]`
4	$[X, 0, Z]$	$\left[0, -x, \frac{1}{4}\right]$	`[4,-8]`
5	$[0, -X, -Z]$	$\left[-x, 0, \frac{3}{4}\right]$	`[9,-14]`
6	$[0, X, -Z]$	$\left[x, 0, \frac{3}{4}\right]$	`[10,-13]`
7	$[X, 0, -Z]$	$\left[0, -x, \frac{3}{4}\right]$	`[11,-15]`
8	$[-X, 0, -Z]$	$\left[0, x, \frac{3}{4}\right]$	`[12,-16]`

Table 44: Wyckoff bond: `8b@8k`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, 0, \frac{1}{4}\right]$	`[1,6]`
2	$[-X, 0, 0]$	$\left[-x, 0, \frac{1}{4}\right]$	`[2,5]`
3	$[0, X, 0]$	$\left[0, x, \frac{1}{4}\right]$	`[3,7]`
4	$[0, -X, 0]$	$\left[0, -x, \frac{1}{4}\right]$	`[4,8]`
5	$[-X, 0, 0]$	$\left[-x, 0, \frac{3}{4}\right]$	`[9,14]`
6	$[X, 0, 0]$	$\left[x, 0, \frac{3}{4}\right]$	`[10,13]`
7	$[0, -X, 0]$	$\left[0, -x, \frac{3}{4}\right]$	`[11,15]`
8	$[0, X, 0]$	$\left[0, x, \frac{3}{4}\right]$	`[12,16]`

Table 45: Wyckoff bond: `16c@8k`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, 0, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, 0, \frac{1}{4}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[0, x, \frac{1}{4}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[0, -x, \frac{1}{4}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, 0, \frac{1}{4}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x, 0, \frac{1}{4}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[0, x, \frac{1}{4}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[0, -x, \frac{1}{4}\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[-x, 0, \frac{3}{4}\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[x, 0, \frac{3}{4}\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[0, -x, \frac{3}{4}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[0, x, \frac{3}{4}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[x, 0, \frac{3}{4}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[-x, 0, \frac{3}{4}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[0, -x, \frac{3}{4}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[0, x, \frac{3}{4}\right]$	`[16]`

* Wyckoff site: `8l`, site symmetry: `.2.`

Table 46: Wyckoff bond: `8a@8l`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, X, Z]$	$\left[x, \frac{1}{2}, \frac{1}{4}\right]$	`[1,-6]`
2	$[0, -X, Z]$	$\left[-x, \frac{1}{2}, \frac{1}{4}\right]$	`[2,-5]`
3	$[-X, 0, Z]$	$\left[\frac{1}{2}, x, \frac{1}{4}\right]$	`[3,-7]`
4	$[X, 0, Z]$	$\left[\frac{1}{2}, -x, \frac{1}{4}\right]$	`[4,-8]`
5	$[0, -X, -Z]$	$\left[-x, \frac{1}{2}, \frac{3}{4}\right]$	`[9,-14]`
6	$[0, X, -Z]$	$\left[x, \frac{1}{2}, \frac{3}{4}\right]$	`[10,-13]`
7	$[X, 0, -Z]$	$\left[\frac{1}{2}, -x, \frac{3}{4}\right]$	`[11,-15]`
8	$[-X, 0, -Z]$	$\left[\frac{1}{2}, x, \frac{3}{4}\right]$	`[12,-16]`

Table 47: Wyckoff bond: `8b@8l`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, \frac{1}{2}, \frac{1}{4}\right]$	`[1,6]`
2	$[-X, 0, 0]$	$\left[-x, \frac{1}{2}, \frac{1}{4}\right]$	`[2,5]`
3	$[0, X, 0]$	$\left[\frac{1}{2}, x, \frac{1}{4}\right]$	`[3,7]`
4	$[0, -X, 0]$	$\left[\frac{1}{2}, -x, \frac{1}{4}\right]$	`[4,8]`
5	$[-X, 0, 0]$	$\left[-x, \frac{1}{2}, \frac{3}{4}\right]$	`[9,14]`
6	$[X, 0, 0]$	$\left[x, \frac{1}{2}, \frac{3}{4}\right]$	`[10,13]`
7	$[0, -X, 0]$	$\left[\frac{1}{2}, -x, \frac{3}{4}\right]$	`[11,15]`
8	$[0, X, 0]$	$\left[\frac{1}{2}, x, \frac{3}{4}\right]$	`[12,16]`

Table 48: Wyckoff bond: `16c@8l`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, \frac{1}{2}, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{2}, \frac{1}{4}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, x, \frac{1}{4}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, -x, \frac{1}{4}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, \frac{1}{2}, \frac{1}{4}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x, \frac{1}{2}, \frac{1}{4}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, x, \frac{1}{4}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, -x, \frac{1}{4}\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[-x, \frac{1}{2}, \frac{3}{4}\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[x, \frac{1}{2}, \frac{3}{4}\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[\frac{1}{2}, -x, \frac{3}{4}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[\frac{1}{2}, x, \frac{3}{4}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[x, \frac{1}{2}, \frac{3}{4}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[-x, \frac{1}{2}, \frac{3}{4}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{1}{2}, -x, \frac{3}{4}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[\frac{1}{2}, x, \frac{3}{4}\right]$	`[16]`

* Wyckoff site: `8m`, site symmetry: `m..`

Table 49: Wyckoff bond: `8a@8m`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[x, y, 0]$	`[1,10]`
2	$[-X, -Y, 0]$	$[-x, -y, 0]$	`[2,9]`
3	$[-Y, X, 0]$	$[-y, x, 0]$	`[3,12]`
4	$[Y, -X, 0]$	$[y, -x, 0]$	`[4,11]`
5	$[-X, Y, 0]$	$\left[-x, y, \frac{1}{2}\right]$	`[5,14]`
6	$[X, -Y, 0]$	$\left[x, -y, \frac{1}{2}\right]$	`[6,13]`
7	$[Y, X, 0]$	$\left[y, x, \frac{1}{2}\right]$	`[7,16]`
8	$[-Y, -X, 0]$	$\left[-y, -x, \frac{1}{2}\right]$	`[8,15]`

Table 50: Wyckoff bond: `8b@8m`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[x, y, 0]$	`[1,-10]`
2	$[0, 0, Z]$	$[-x, -y, 0]$	`[2,-9]`
3	$[0, 0, Z]$	$[-y, x, 0]$	`[3,-12]`
4	$[0, 0, Z]$	$[y, -x, 0]$	`[4,-11]`
5	$[0, 0, -Z]$	$\left[-x, y, \frac{1}{2}\right]$	`[5,-14]`
6	$[0, 0, -Z]$	$\left[x, -y, \frac{1}{2}\right]$	`[6,-13]`
7	$[0, 0, -Z]$	$\left[y, x, \frac{1}{2}\right]$	`[7,-16]`
8	$[0, 0, -Z]$	$\left[-y, -x, \frac{1}{2}\right]$	`[8,-15]`

Table 51: Wyckoff bond: `16c@8m`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -y, 0]$	`[2]`
3	$[-Y, X, Z]$	$[-y, x, 0]$	`[3]`
4	$[Y, -X, Z]$	$[y, -x, 0]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, y, \frac{1}{2}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x, -y, \frac{1}{2}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[y, x, \frac{1}{2}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[-y, -x, \frac{1}{2}\right]$	`[8]`
9	$[-X, -Y, -Z]$	$[-x, -y, 0]$	`[9]`
10	$[X, Y, -Z]$	$[x, y, 0]$	`[10]`
11	$[Y, -X, -Z]$	$[y, -x, 0]$	`[11]`
12	$[-Y, X, -Z]$	$[-y, x, 0]$	`[12]`
13	$[X, -Y, Z]$	$\left[x, -y, \frac{1}{2}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[-x, y, \frac{1}{2}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[-y, -x, \frac{1}{2}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[y, x, \frac{1}{2}\right]$	`[16]`

* Wyckoff site: `16n`, site symmetry: `1`

Table 52: Wyckoff bond: `16a@16n`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -y, z]$	`[2]`
3	$[-Y, X, Z]$	$[-y, x, z]$	`[3]`
4	$[Y, -X, Z]$	$[y, -x, z]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, y, \frac{1}{2} - z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x, -y, \frac{1}{2} - z\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[y, x, \frac{1}{2} - z\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[-y, -x, \frac{1}{2} - z\right]$	`[8]`
9	$[-X, -Y, -Z]$	$[-x, -y, -z]$	`[9]`
10	$[X, Y, -Z]$	$[x, y, -z]$	`[10]`
11	$[Y, -X, -Z]$	$[y, -x, -z]$	`[11]`
12	$[-Y, X, -Z]$	$[-y, x, -z]$	`[12]`
13	$[X, -Y, Z]$	$\left[x, -y, z + \frac{1}{2}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[-x, y, z + \frac{1}{2}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[-y, -x, z + \frac{1}{2}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[y, x, z + \frac{1}{2}\right]$	`[16]`